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REVIEW ARTICLE

# Brahmi (*Bacopa monnieri)*: an ayurvedic herb in the management of various diseases

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## ABSTRACT

*Bacopa monnieri* (Scrophulariaceae) is an annual herb, widely distributed throughout the Indian subcontinent. It is known as Brahmi. The whole plant is extensively used in almost all traditional systems of medicine in India. It is a tiny, slender, succulent herb. The creeping stems that grow leaves and flowers have nodes that generate roots, and their length ranges from 10 to 30 cmsoft and 1-2 mm thick. The flowers are a light blue or pinkish-white colour. Triterpenoids, saponins, and bacosides havebeen identified to be this plant's major chemical therapeutic components through various studies. It has been found that the main element causing therapeutic benefits is bacoside A. It is used as an anti-inflammatory, endocrine, gastrointestinal, antimicrobial, smooth muscle relaxant effects, analgesic, anticancer, antidiabetic, antifungal, cardiovascular, anti-Ischemic, anti-emetic, anti-epilepticand antioxidant but it is best known as a memory enhancer. This review focuses on the phytochemical constituents present in *B. monnieri*, its multifarious pharmacological activities, and clinical studies.

**Keywords:** *Bacopa monnieri*, bacosides, bacogenins, bacopasaponins, multifarious pharmacological activities

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## INTRODUCTION

*Bacopa monnieri* (Linn.) Pennell. Syn. *Herpestis monnieri* L. family Scrophulariaceae, named Brahmi in Hindi is an annual or perennial herb that grows in marshy areas of India and climbs to an altitude of 1,320 meters (Kritikar and Basu, 1987). It is glabrous and somewhat succulent. Most ancient Indian physicians and Ayurvedic practitioners utilized herbs for treating dermatitis, anemia, and diabetes. It is also used as a safe cardiac tonic(Chopra et al., 1956; Asvikar et al., 1992). The entire plant has a strong reputation in Indian medicine for being a nervine tonic and a treatment for epilepsy and mania (Wealth of India 1998; Jain, 1991).

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*B. monnieri* has stems 10 to 30 cm. soft and 1-2 mm thick. Sessile, obovate-oblong or spatulate, entire obscure, bottom surface spotted, leaves measure between 0.6 and 2.5 cm long and 3 to 8 mm wide. Flowers are axillary and solitary, blue or white with purple veins, up to 5 mm long ovoid, glabrous capsules, and seeds up to 0.85 mm long, oblong, striate, and pale. It tastes slightly bitter and has no strong aroma.

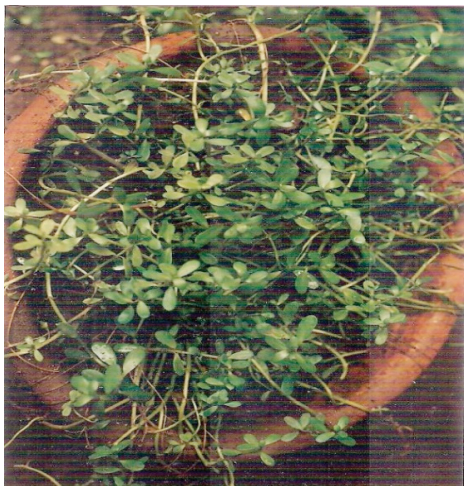


Figure 1:*Bacopa monnieri* (Tamboli et al., 2018)

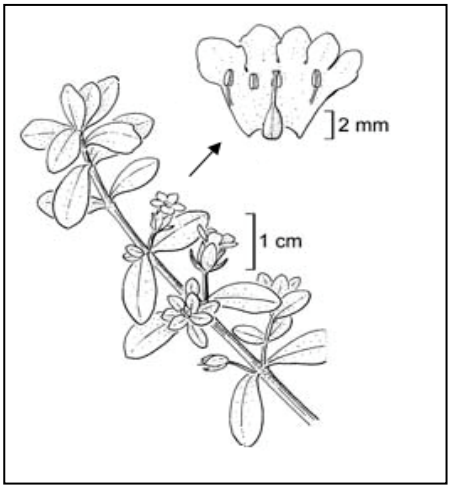



Figure 2: Morphological features of Brahmi

**PHYTOCHEMISTRY**

For its phytochemical components and numerous pharmacological research, *B. monnieri* has been widely used. A triterpenoid saponin, bacoside A (mp 250$^{o}$C), and bacoside B (mp 203$^{o}$C) were identified from *B. monnieri* (Chatterji et al., 1963). Bacoside A was obtained in the free state and bacoside B was found in the form of mixed crystals with bacoside A (Chatterji et al.,1965). On acid hydrolysis bacoside A gives two sugars i.e. D-glucopyranose, L-arabinopyranose, and a disaccharide which on hydrolysis gives glucose, and arabinose. The bacoside B on acid hydrolysis gives A1, A2, A3, and A4 bacogenins, glucose, arabinose (Basu et al.,1967). He further reported bacoside B as dextrorotatory and bacoside A as levorotatory compounds. The former has twice the hemolytic activity of the latter, which appears to be related to the former's different carbohydrate chain structure. Then the structure of bacogenin A1 (mp 242$^{o}$C) and bacogenin A2 were confirmed. It was discovered that bacogenin A2 was an isomer of bacogenin A1, with the two being distinct either in their structure at the C-20 position or in the position of the noncyclic methyl group side chain. (Kulshrestha et al.,1947; Kulshrestha et al.,1974; Kulshrestha et al.,1975). The structure of bacogenin A3 was also confirmed from this plant (Chandel et al.,1975).

The structure of bacogenin A4 as ebeline lactone was reported (Kulshreshtha et al., 1973). Then the new minor triterpenoid saponin bacoside A1[6] have isolated from the plant (Jain and Kulshreshtha, 1993). The bacoside A2 from *B. monnieri* and elucidated it as 3 β- [-O-α-L-arabino furanosyl (1→6)-O-[α-L-arabinopyranasy-1(1→5)]-O-α-D-gucofuranosy-1) oxy] pseudojujubogenin was reported (Rastogiand Kulshreshtha,1993). Similarly, in another study a new triterpenoid saponin bacoside A3 was isolated (Rastogi et al.,1994). Then in another study jujobogenin and pseudojujubogenin the degradative products of bacoside A by Smith-de Mayo degradation reaction were reported (Ken-Ichi et al., 1974; Ken-Ichi et al.,1978). Then three novel triterpenoid saponins were discovered that are dammarane class, bacopasaponins B, bacopasaponins C

(Garai et al., 1996). The two new dammarane-type jujubogeninbisdesmosides. Bacopasaponins E and F and defined them as 3-O-[ β-D-glucopyranosy (1→3)-{α-L-arabinofururanosyl (1→2)} α-L arabinopyranasy 1) jujobogenin and 3-O-[β-D-glucopyranosyl (1→3) { α-L-arabinofururanosyl (1→2) } β-D- glucopuranosyl]-20-O-L-arabinopyranosyl] jujubogenin respectively (Mahato et al., 2000). Another study reports the isolation and characterization of bacopaside I and II. (Chakravorty et al., 2001). The phytochemical investigation of aerial parts of *B. monnieri* has led to the discovery of a novel triterpene, lup-20, as well as the sterols 13,14,sfeco-stigma-9 (11)-14, and 15-dien-3 (2a) Becosine is the name for the -en-3-ol-2-oic acid (BaharAhmed and Rahaman A, 2000). A new neutral saponin, monnierin, and was shown to be a triterpenoid saponin which on acid hydrolysis affords glucose and arabinose (Dutta and Basu 1963, Dutta and Basu 1967, Dutta and Basu 1968). Then the compound was isolated as nicotine and 3-formyl -4hydroxy-24-pyran from the plant ( Schulle et al, 1972).

Some other constituents have been reported such as betulic acid, D-mannitol, sigmastanol, β-sitosterol, and sigmasterol along with another major constituent of the plant. A chemically pure saponin named as (hersaponin) (mp. 223-234$^0$C) was isolated in addition to D-mannitol and potassium salts. Aspartic acid, Glutamic acid, Serine, and A-alanine have also been reported (Sastry et al., 1959). Two alkaloids viz. herpenstine and brahmine were reported from this plant (Brown and Martin, 1960). The structures of various constituents of *B. monnieri* have been given in Table 2. In the nitro blue tetrazolium (NBT) experiment, the hydroalcoholic extract of *B. monnieri* released from polymorphonuclear (PMN) cells revealed that the main dammarane triterpene saponin bacoside A3 was in charge of this effect in herbs. The ability of *B. monnieri* node, internode, and leaf explants to produce new organs was investigated (Tiwari et al.,1998).

**ISOLATION**

Isolation of Bacosides and other fractions from the *B. monnieri*

Following is the general process for isolating bacosides and other fractions from *B. monnieri* extracts:

1. Extract powdered drug *B. monnieri* Exhaustively with ethanol,
2. Concentrate ethanolic extract by dissolving it in 60% ethanol,
3. Use a separating funnel and benzene to extract the above solution to produce,
    a. Ethanol Extract (X), b) Benzene Extract (Y)
4. A thick slimy substance precipitates out of the ethanolic extract (X) during concentration, separating it from the supernatant. (Z)
5. To separate the brown resinous ingredient, dissolve the slimy mass in alcohol and fractionally precipitate it with ether and petroleum ether several times.
6. Concentrate the mixture of ethanol, ether, and petroleum ether; macerate the leftover material with acetone, and filter it.
7. Divide the insoluble powder in acetone between butanol and water.
8. A precipitate settles down while the butanol soluble fraction is being concentrated. Filter it, then mark it as FI.
9. Concentrate the filter once more, and then use acetone and ether to precipitate it into a powder.
10. Dry out the mother liquor after concentration. It provides a brown powder FIII.
11. When a solid settle down, keep the acetone-soluble fraction in the cold for a few days. Filter it and acetone-wash it FIV.
12. Macerate the hygroscopic, resinous substance with ethyl acetate after almost completely drying the filtrate. Benzene solution Y should get the solvent washings.

Bacoside A is produced in the F1, F2, and FIII fractions by the aforementioned isolation process, whereas Bacoside B is produced in the FIV fraction.

The paper chromatographic study of the solvent system with FI, FII, FIII, and FIV. For the first three fractions of bacoside A, ethyl acetate-pyridine-water (4:1:1) and treatment with the developing reagent trichloroacetic acid (25%) in chloroform result in Rf values of 0.43 and 0.09, respectively (Chatterji et al., 1963).

## ASSAY / ANALYTICAL STUDIES

UV spectroscopy for the quantitative identification of all bacosides as bacoside A

Make a 1 mg/mL solution of bacoside A in methanol. Prepare additional dilutions of this stock solution with concentrations of 25, 50, 100, 200, 800, and 1000 mg/mL. Add 1ml of 4 N aqueous sulphuric acid to 1ml of each of the aforementioned solutions, and reflux for 4 hours on a water bath. Under reduced pressure, the methanol is extracted after cooling and dilution with 4 mL of distilled water. Chloroform (4x5ml) is used to extract the aqueous solutions. The combined chloroform extract is then washed with a 0.1% ammonia solution before being rinsed with distilled water. Remove the chloroform at reduced pressure after drying the chloroform solution on anhydrous sodium sulphate. Make a solution of the residue in 10 ml of AR-grade methanol. Draw a calibration curve (n=5) based on the solutions' absorbance at 278 nm following hydrolysis, with the range of 5 to 50 g/ml representing bacoside A.

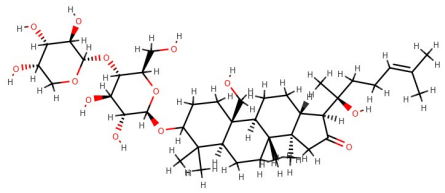
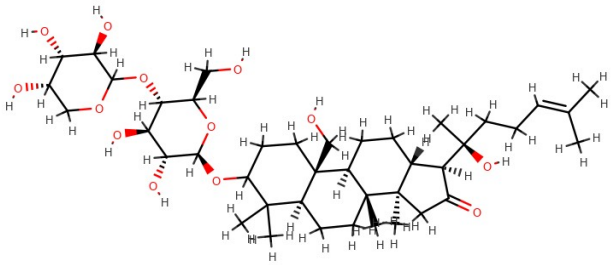
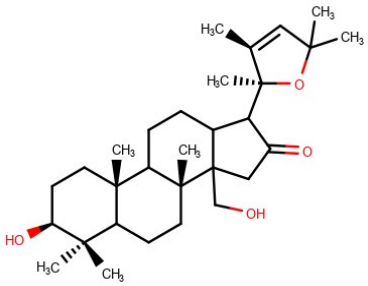
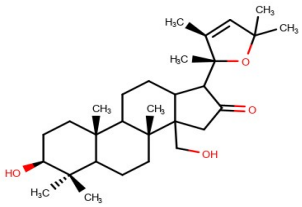
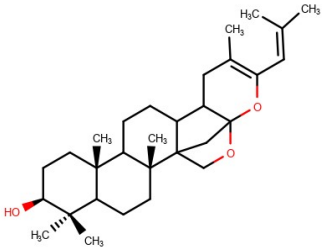
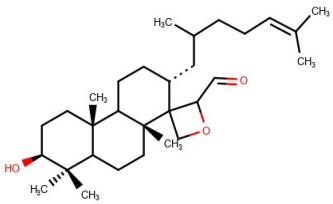
Test Solution: To create a solution with a concentration of roughly 300 g/ml, dissolve a known amount of the sample (bacosides) in methanol. 1 ml of the test solution should be hydrolyzed before proceeding as directed under the calibration curve. Calculate the sample's bacoside A content using the calibration curve and the absorbance at 278 nm. Table 2 shows the average proportion of the different *B. monnieri* as elements as determined by the aforementioned procedure (Indian Herbal Pharmacopoeia 1998).

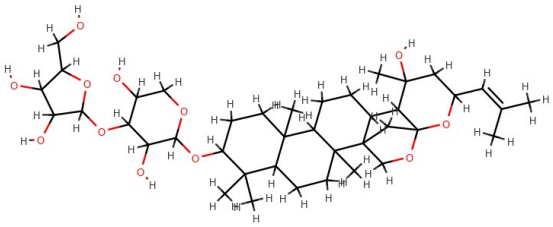
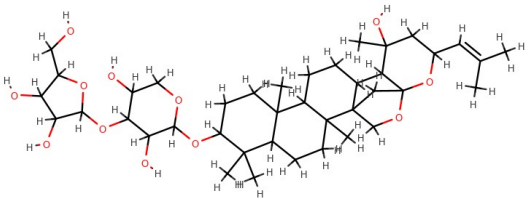
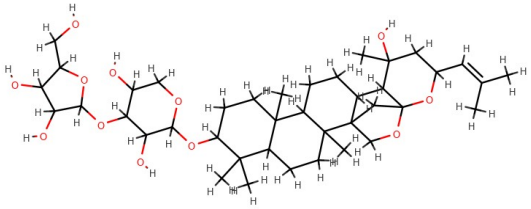
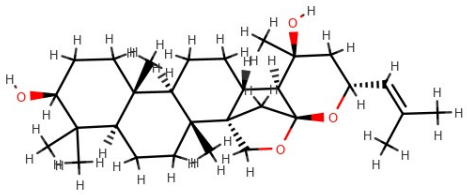
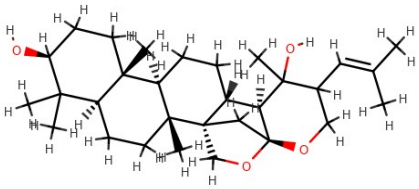
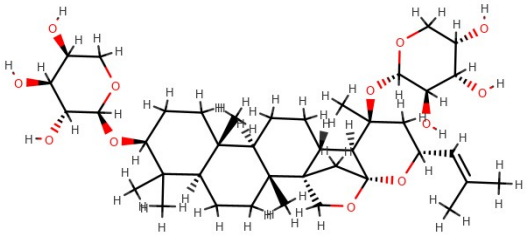
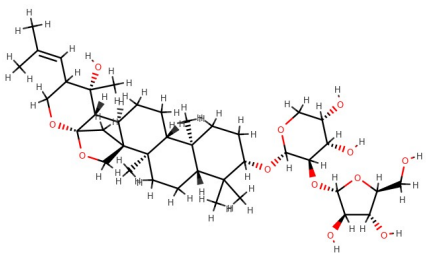
**Table 1: Bacosides and other constituents isolated from *B. monnieri***

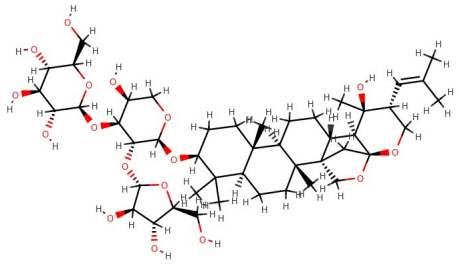
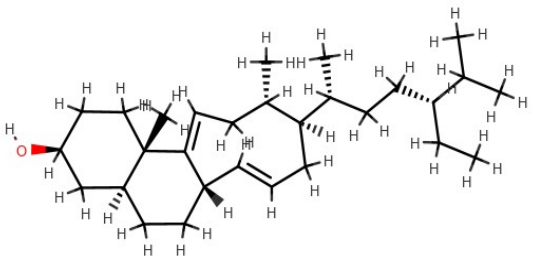
| Compound | m.p. ($^{O}$C) | % Yield (Dry weight basis) |
|---|---|---|
| **Bacoside A** | 250-51(dec.*) | 1.54 |
| **Bacoside B** | 230 (dec.*) | 0.65 |
| **Betulic acid** | 315 | 0.11 |
| **D- Mannitol** | 166 | 0.02 |
| **Stigmasterol** | 170 | 0.013 |
| **B-Sitosterol** | 137 | 0.014 |
| **Stigmastanol** | 147 | 0.005 |

*decomposition

**Table 2: Phytoconstituents of *Bacopa monnieri***

| Phytoconstituents | Structures |
|---|---|
| Bacoside A |  |
| Bacoside B |  |
| Bacogenin $A_1$ |  |
| Bacogenin $A_2$ |  |
| Bacogenin $A_3$ |  |
| Bacogenin $A_4$ |  |

| | |
|---|---|
| Bacoside $A_1$ |  |
| Bacoside $A_2$ |  |
| Bacoside $A_3$ |  |
| Jujobogenin |  |
| Pseudojujubogenin |  |
| Bacopasaponins A |  |
| Bacopasaponins B |  |

| | |
|---|---|
| Bacopasaponins C |  |
| Bacosterol |  |

## PHARMACOLOGY

The herb *B. monnieri* possesses a variety of pharmacological activities. The plant has a particularly good reputation in Indian medicine as a laxative, carminative, digestive, anti-inflammatory, anti-convulsant, cardiotonic, bronchodilator, diuretic, febrifuge, and nervine tonic. It helps with vitiated disorders of the Kapha and Vata, as well as with epilepsy, dementia, tumors, ulcers, dyspepsia, constipation, asthma, bronchitis, skin ailments, leprosy, leukoderma, syphilis, dysmenorrhea, sterility, fever, and general weakness (Asvikar et al., 1992, Dandiya 1992, Indian Medicinal Plants, 1994, Jain et al., 1991, Nadkarni, 1982).

**Nervine Tonic**: The herb has been utilized as a nervine tonic for memory enhancement since the dawn of time. Even though it has gotten a lot of attention in Indian works on religion, culture, and medicine. The first time the CNS effects of Brahmi are mentioned directly is in the Charak Samhita, which was composed in the first century AD. Brahmi is a remedy for mental diseases, which include symptoms like anxiety, difficulty concentrating, intellectual weakness, and others, according to Charak. Sushrut Samhita, another reliable treatise, describes brahmi as a useful medication in cases of mental and memory loss ( Singh and Dhawan, 1977, Vaidya Ashok, 1997).

The effectiveness of *B. monnieri* and chlorpromazine on rats ability to learn a "basic motor learning test" has been compared and evaluated. When both medications are given to animals who are still learning, their performance is improved. At a dose of 10 mg/kg, chlorpromazine seemed to depressive effect on motor efficiency as seen by the comparatively long time some animals took to complete the T-path, while Brahmi had no such negative effects. Bhardwaj and colleagues found that Brahmi, one of the components of CIHP (III), was successful in enhancing mental capacity. When CIHP (III) is administered to albino rats, avoidance learning during endurance performance significantly improves. BR-16 A (Mentat8). According to reports, a natural psychotropic medication has a positive impact on people with mental retardation, cerebral deficits, and behavioral disorders (Kulkarni and Verma, 1993).

**Depressant Activity**: In their research of the crude total extract of *B. monnieri*, found that the degree of neurological deficit in rats varied dose-dependently and was depicted by abnormalities in sitting posture and gait, a loss of muscle tone, an inability to maintain balance, and the absence of a righting reflex. It also produced a slight to considerable respiratory depression. On

the nerve-muscle preparation of the frog's gastrocnemius muscle, the effect of the crude whole extract at concentrations ranging from 1:00,000 to 1:1000 was investigated. Each concentration was tested in six to ten experiments, with the frog's other gastrocnemius muscle acting as the control in each experiment. The crude extract had no impact at concentrations of 1:100,000 to 1:20,000. While it generated total neuromuscular block at concentrations of 1:5,000 to 1:1,000 and partial neuromuscular block at 1:10,000, the muscle was still sensitive to direct stimulation. Rats and guinea pigs were found to be anesthetized by the pure saponin glycoside compound (Malhotra and Das, 1959). According to research done the entire extract and its components have an inhibitory effect on the rat brain's homogenate's ability to take in oxygen (Dhalla et al.,1963).

**Anxiolytic Activity:** When rats were treated with the plant showed signs of reduced anxiety. The extract significantly decreased the activity of experimental animals without significantly changing the brain's neurochemistry, and the substance significantly enhanced the barbiturate hypnotic effect (Singh et al., 1979). A dose-dependent anxiolytic effect similar to lorazepam has been established by extract of *B. monnieri* that contains 25.5% bacoside A. The results are consistent with the plant's clinical use for the improvement of cognitive functioning in the Indian medical system (Bhattacharya and Ghosal,1998).

**In Estrogen-deficiency:** For the treatment of estrogen-deficiency-related cognitive impairment, a multi-component herbal preparation comprising 15–25% *B. monnieri* extract is employed. Menopausal women who showed signs of an estrogen insufficiency disease were successfully treated with the herbal combination (Grose, 2001).

**Bronchodilatory Activity:** An ethanolic extract of the herb had bronchodilatory and circulatory effects on rats under anesthesia. The extract countered carbachol's bronchoconstrictor effects. When compared to isoprenaline, the plant extract's bronchodilatory impact was manifested by a decrease in expiratory pressure that was more similar to a salbutamol-induced effect. The bronchodilatory effect of the plant extract is probably jointly mediated by independent and adrenoceptor-dependent mechanisms. Consequently, providing support for its traditional use in the treatment of asthma (Dar and Channa, 1977).

**Anti-inflammatory Activity:** The drug's anti-inflammatory efficacy is mediated via interfering with prostaglandin synthesis and/or action, as well as maybe by stabilizing lysosomal membranes. It has similar anti-inflammatory properties to indomethacin (Khanna, 1995).

**Antistress Activity:** The effect of four MedhyaRasayana drugs viz. Brahmi (*B. monnieri*) Shankhapushpi (*Convolvulus pluricaulis*) Yasti (*Glycyrrhiza glabra*) and Guduchi (*Tinospora cardifolia*) have been studied. In normal and stressed rats, the brain's levels of acetylcholine, catecholamines, 5-hydroxytryptamine, and histamine were examined. These medications significantly increased acetylcholine levels in the cortex while significantly lowering acetylcholine levels in the homogenate of the entire brain. Additionally, observed to be higher were the catecholamine and 5-hydroxy tryptamine concentrations. Rats under stress had more noticeable changes. The aforementioned alterations suggested that these medications, in addition to working as tranquilizers, may also be improving mental performance (Singh et al., 1979).

**Anticancer Activity:** There have been claims that plant extract possesses anti-cancer qualities. Sarcoma 180 cell culture was used to evaluate various doses of the ethanolic extract of *B. monnieri*. We evaluated the DNA incorporation of 3, 4-thymidine and the viability of cell growth. With increasing *B. monnieri* extract concentration, Sarcoma 180 cell proliferation was severely suppressed. A DNA replication stage has been proposed as the drug's site of action for its anticancer effect (Elangovan et al., 1995). The rasayanas that produced Madhur vipaka (*B. monnieri* and other crude drugs) improved the carbon clearance in rats, indicating stimulation of the reticuloendothelial system (Dahanukar and Thatte, 1977).

**Anticonvulsant Activity:** The anticonvulsant profile of several extracts of *Centella asiatica* and *B. monnieri* in rats has been assessed. At 34 and 64, *B. monnieri* extracts (500 mg/kg) exhibited mild to moderate anticonvulsant action, whereas 1000 mg/kg showed low activity. At 1 and 6 hours, there was noticeably more activity, but not at 24 hours. *B. monnieri* demonstrated a quicker beginning of the action, and time/dose response was comparable to the impact of phenytoin, a common medicine (Sudha et al., 2002).

**Analgesic Activity:** Bacosine has been found to have analgesic properties. It has a little analgesic effect, but it has no impact on catalepsy brought on by haloperidol, barbiturate narcosis, spontaneous motor activity, or conditioned avoidance reaction. According to reports, the analgesic action is opioidergic in nature (Vohora and Khanna, 1977).

**MAO Inhibitor:** The rat liver and brain stem MAO enzymes were both activated by the aqueous extract of *B. monnieri* reaching a maximum at 20 mcg/ml of the reaction mixture at lower concentrations. The action decreases as the dry concentration rises, and it becomes an inhibitor at 30-35 mcg/ml RM. MAO's greatest inhibition in this investigation came from the brain stem (Ramakrishnan, 1987). A cosmetic lotion containing *B. monnieri* extract 3% has been reported to be advantageously used in cases of stress, whether physical-chemical, or psychological, which affects skin metabolism and skin function and appearance (Linter, 1999). The superoxide inhibiting effect of bacoside. In the nitroblue tetrazolium (NBT) experiment, the hydroalcoholic extract of *B. monnieri* has released from polymorphonuclear (PNM) cells. It was discovered that the main dammarane triterpene saponin bacoside A3 is what produces these effects in herbs (Pawar et al., 2001).

**Anti-Diabetic Activity:** A metabolic condition known as diabetes mellitus affects the metabolism of carbohydrates, fats, and proteins. This disease affects 1% of the population, on average. Diabetes mellitus is a serious issue (Sabina et al., 2014). In their study of an ethanolic extract of *B. monnieri*, revealed that the triterpene bacosine seems to be what causes the increased glycogen content in diabetic rats (Ghosh et al., 2011). Additionally, the extract resulted in an increase in peripheral glucose consumption in the diaphragm of diabetic rats in vitro. Consequently, alloxan-induced diabetic rats revealed insulin-like action (Sabina et al., 2014).

**Anti-Microbial:** Researchers examined the anti-microbial properties of ethanolic, diethyl acetate, ethyl acetate, and aqueous extracts of *B. monnieri* aerial parts (Sampathkumar et al., 2008). While diethyl ether extract indicated antibacterial activity against gram-positive organisms, ethyl acetate extract showed antibacterial action against gram-negative species. In the test, bacteria including Proteus vulgaris and Staphylococcus aureus were used. The ethanolic extract showed anti-fungal action against *Candida albicans* and *Aspergillus niger*. *B. monnieri* extracts in ethyl acetate and methanol were shown to have antibacterial activity by the disc diffusion method against 7-gram-negative and 11-gram-positive bacteria (Khan et al., 2010). Two phytochemicals from *Bacopa monnieri's* aerial parts, betulinic acid and orexin, have been discovered to have antifungal effects on *Alternaria alternata* and *Fusarium fusiformis* (Sampathkumar et al., 2008).

**Cardiovascular:** The effects of intravenous *B. monnieri* extract administered to rats under anaesthesia on arterial blood pressure and heart rate. Without altering heart rate, the extract reduced systolic and diastolic pressures. Blood pressure was shown to decrease as a result of processes affecting vascular smooth muscle Ca2+ homeostasis and the endothelium's release of nitric oxide (Kamkaew et al., 2011).

**Antifungal Activity:** The IC50 values for Fusarium oxysporum (31.25 g/ml), Sclerotium rolfsii (6.25 g/ml), Alternaria sp (28.75 g/ml), and Rhizoctonia solani (18.75 g/ml) show that extract of bacopa successfully inhibited the development of the selected fungus (Jain et al., 2017).

**Anti-emetic Properties:** Research on *B. monnieri's* anti-emetic effects was done. Chemotherapy-induced emesis in Suncus murinus was used to examine the anti-emetic effects of *B. monnieri's* methanolic and butanoic fractions. The animal was administered an injection of cisplatin (30 mg/kg) to cause emesis. For anti-emetic screening, different concentrations of butanoic extract (5-20 mg/kg) and methanolic extract (10-40 mg/kg) were assessed. The results showed that methanolic extract inhibited the vomiting reaction by 71% and butanoic extract by 59.4%. It promotes methanolic extract as a reliable source of medication to treat nausea caused by chemotherapy (Ullah et al., 2017).

**Anti-Rheumatic Activity:** Along with significantly increased paw edema and specific signs of arthritis, arthritic index, neutrophil infiltration, higher anti-collagen IgM and IgG serum levels, and inflammatory mediators including cyclooxygenase, lipoxygenase, and myeloperoxidase were also observed. IgM and IgG anti-collagen levels in serum were diminished (Viji et al., 2010).

**Antiparkinsonian activity:** *Bacopa monnieri* has shown its activity as a potential anti-Parkinsonian treatment by reducing alpha-synuclein aggregation in Caenorhabditis worms' models of Parkinson's disease, which prevents the dopaminergic neurodegeneration, and restores the lipid content in nematodes (Jadiya et al., 2012).

**Anti-Ulcerative activity:** *Bacopa monnieri*juice was administered orally twice daily for five days at doses of 100 and 300 mg/Kg, along with 250 mg/Kg of sucralfate. Except for ethanol-induced ulcers, the juice exhibited antiulcer efficacy in models of stomach ulcers. The outcome is associated to aspects that protect the mucosa, such as elevated mucin secretion, reduced mucosal cell exfoliation, and mucosal glycoprotein (Mohanty et al., 2010).

**Anti-Ischemic Activity:** The 48 rats were split into four groups at random: the sham group, the *Bacopa monnieri* control group, the ischemia-reperfusion control group, and the *Bacopa monnieri*-treated group. Significant myocardial necrosis, apoptosis, heart rate lowering, a decline in antioxidant status, and an increase in lipid peroxidation were all effects of post-ischemic reperfusion injury (Mathur et al., 2010).

**Antioxidant Activity:** Four *B. monnieri* (whole plant) extracts' antioxidant potential was examined using DPPH radical scavenging. The methanol and aqueous sequential extracts had the highest antioxidant activity, with IC.50 values of 46.00 g/ml and 43.10 g/ml, respectively (Kar et al., 2002).

**Endocrine effects:** When administered orally to mice, bacopa extract (200 mg/kg) increased the thyroid hormone T4 by 41%. T4 was not converted to T3 by the extract, indicating that it may instead directly stimulate T4 production and/or release at the glandular level (Shahid and Subhan, 2014).

**Hepatoprotective Activity:** Methanolic extract pre-treatment against CCl4 has revealed a protective effect. In addition to congesting the central vein and enlarging the sinusoids, it has preserved the liver structure in rats. The screening of hepatoprotective herbal medicines has frequently utilized chemically induced hepatotoxicity in animals (Shahid and Subhan, 2014).

**Anti-Alzheimer Activity:** The memory facilitating effect of *B. monnieri* extract and its active constituents was reported due to an effect on central cholinergic modulation of memory functions (Bhattacharya, 2001).

**Mast-Cell Stabilizing Activity:** *Bacopa monnier i*extracts in successive levels of petroleum ether, chloroform, methanol, and water were examined (in vitro) for their ability to stabilize mast cells. The well-known mast cell stabilizer disodium cromoglycate and the methanolic fraction were equivalent in terms of the robust activity of the fraction (Samiulla, 2001).

**Hair growth promoting activity:** The *Emblica officinalis*, *Bacopa monnierii*, and *Cyperus rotundus* alcoholic extract or whole drug formulation provided the best results when compared to the other formulations, as shown by an increase in follicular size and a lengthening of the anagen phase (Dass, 2015, Jain, 2016, Jain, 2012).

## CLINICAL STUDIES

### Anti - Alzheimer's activity of *Bacopa monnieri* in Human Clinical Trail

Prabhakar et al. conducted a randomized double-blind parallel phase 2b research to compare the effectiveness of *Bacopa monnieri* and the industry-standard medication Donepezil. In the trial, which included 48 patients, 300 mg of *Bacopa monnieri* extract and 10 mg of dopenzal were administered. The Postgraduate Institute (PGI) memory scale and the Alzheimer's Disease Assessment Scale (ADAS-cog) were used to analyze the results. *Bacopa monnieri* is a powerful anti-Alzheimer agent, and studies comparing it to dopenzil have not revealed any significant differences (Prabhakar et al., 2020).

**Effect on Cognitive Function:** Animal studies have shown that *B. monnieri* extracts have a positive impact on cognitive function. The long-term effects of *B. monnieri* extract on cognitive performance in healthy human volunteers have been studied. The individuals were divided into two treatment groups at random: *B. monnieri* (300mg) or placebo in this double-blind, placebo-controlled, independent group trial. Neuropsychological testing of the patients at baseline, 5, and 12 weeks after pharmaceutical administration has showed significantly accelerated visual information processing speed as measured by IT task learning rate and memory consolidation, with the greatest advantages becoming apparent after 12 weeks. These findings suggest that *B. monnieri* may improve higher order cognitive abilities like learning and memory, which are very dependent on environmental input (Stough et al., 2001).

**Effect on Schizophrenia:** Effect on Smriti Sagarovasa, a herbomineral preparation containing Brahmi (*B. monnieri*), *Acorus calamus* and *Celastrus paniculatas* in the dosage of 250 mg tablet given 3 times a day with honey has been reported. Eleven patients out of 30 (36.67%) showed significant improvement and an approximately similar number (33.33%) showed, moderate improvement after 3 months of treatment (Tripathi and Singh, 1994).

**Anti-epileptic Activity:** For 6 weeks, 31 adult epilepsy patients between the ages of 23 and 42 were administered Mentat tablet formulation b.i.d. in addition to the other epileptic medications. Ayurvedic components identified in Mentat pills, including *B. monnieri, Centella asiatica, Whitania somnifera*, *Evolvulus alsinosides*, and *Nordostachys jatamanasi*, significantly reduced seizure frequency and functioned as an effective adjuvant to frequently prescribed anti-epileptic medications. No negative effects were noticed (Maharana and Maharana, 1994)

**Anabolic Activity:** A preparation containing extracts of *B. monnieri* along with some other crude drugs like *Whitania somnifera, Emblica officinalis, Asparagus lacemosus, Tinospora cardifola, Tribulus terrestris, Picorrhiza kuroa, cyperus scariosus, Terminalia chebula, Piper longum*, and *Myristica fragrans* when administered to patients with complaints to lethargy, fatigue and exhaustion showed overall improvement. The serum proteins and hemoglobin levels have also shown a significant increase suggesting the anabolic activity of the formulation (Jayaram et al., 1993).

**Anti-diabetic Activity**: In diabetic patients, clinical trials of the Rasayana medicines, *Centella asiatica, Asparagus racemosus*, and *B. monnieri* were conducted 78. Serum IgA levels were found to be considerably lower in diabetics, higher, and comparable to IgM levels. Serum IgA levels in diabetic patients were reduced by all medications. However, *A. racemosus* was the only species with statistically significant results (Arora et al., 2002)

## CONCLUSION

Since traditional medicines are less hazardous, it is clear that the needs of the great majority of people with health issues are not being adequately met by currently available treatments. Bacopa is one of the indigenous plants used in Ayurveda and herbal remedies. Through preclinical, clinical testing and evaluation in various disease states, the therapeutic efficacy of *Bacopa monnieri* has been established and is widely used in Indian medicine.

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